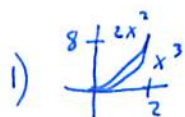


1352 Final Exam Key



a) $Area = \int_0^2 (2x^2 - x^3) dx = \int_0^8 \sqrt[3]{y} - \sqrt{\frac{y}{2}} dy$

b) $Volume = \int_0^2 \pi (2x^2 - x^3) dx = \int_0^8 \pi \left((\sqrt[3]{y})^2 - \left(\sqrt{\frac{y}{2}} \right)^2 \right) dy$

c) $Volume = \int_0^2 \pi (2x^2 + 3)^2 - \pi (x^2 + 3)^2 dx = \int_0^8 2\pi(y+3) \left(\sqrt[3]{y} - \sqrt{\frac{y}{2}} \right) dy$

d) $M_x = \int_0^2 \left(\frac{2x^2 + x^3}{2} \right) (2x^2 - x^3) dx$



$Area = \int_0^{\pi/2} \frac{1}{2} (3 \sin 2\theta)^2 d\theta = \frac{9}{2} \int_0^{\pi/2} \frac{1}{2} (1 - \cos 4\theta) d\theta = \frac{9}{4} \left(\theta - \frac{1}{4} \sin(4\theta) \right) \Big|_0^{\pi/2} = \frac{9\pi}{8}$



a) $\int x^2 \cos(2x) dx = x^2 \left(\frac{1}{2} \sin 2x \right) - \int 2x \left(\frac{1}{2} \sin 2x \right) dx = \frac{1}{2} x^2 \sin(2x) - \left(\frac{1}{2} x \cos 2x - \int \frac{1}{2} \cos 2x dx \right)$
 $= \frac{1}{2} x^2 \sin(2x) + \frac{1}{2} x \cos(2x) - \frac{1}{4} \sin(2x) + C$

b) $\int \frac{3x-2}{x(x+2)} dx = \int \frac{-1}{x} + \frac{4}{x+2} dx = -\ln|x| + 4\ln|x+2| + C$

c) Let $x = 3 \sin \theta$

 $\int \frac{dx}{x^2 \sqrt{9-x^2}} = \int \frac{3 \cos \theta d\theta}{(3 \sin \theta)^2 (3 \cos \theta)} = \frac{1}{9} \int \csc^2 \theta d\theta = \frac{-1}{9} \cot \theta + C$
 $= \frac{-1}{9} \frac{\sqrt{9-x^2}}{x} + C$

d) Let $u = x^3 + 1$
 $du = 3x^2 dx$
 $\int 2x^2 (x^3 + 1)^5 dx = \frac{2}{3} \int u^5 du = \frac{2}{3} \frac{u^6}{6} + C = \frac{2}{18} (x^3 + 1)^6 + C$

4) $\int_2^{\infty} \frac{dx}{x(3+\ln x)^2} = \lim_{N \rightarrow \infty} \int_2^N \frac{dx}{x(3+\ln x)^2} = \lim_{N \rightarrow \infty} \int_{3+\ln 2}^{3+\ln N} \frac{du}{u^2} = \lim_{N \rightarrow \infty} \left(\frac{-1}{3+\ln N} - \frac{-1}{3+\ln 2} \right) = \frac{1}{3+\ln 2}$

5) a) $\sum \frac{k+3}{2^k}$ Ratio or Root Test $\rightarrow \frac{1}{2} \therefore$ conv.

b) LCT w/ $\sum \frac{1}{k^2}$ or $\sum \frac{3}{k(k+1)} = \sum \frac{-3}{k} + \frac{3}{k+1}$ telescoping $\therefore$ conv.

c) $\sum \frac{k^k}{5k-1}$ div. by D.V. Test. Also Ratio, Root, and DCT

b) $\sum \frac{(-1)^k \ln k}{k}$ conv. conditionally b/c $\lim_{k \rightarrow \infty} \frac{\ln k}{k} = 0$, but $\sum \frac{\ln k}{k}$ d.l. by Int. Test DCT log-q Series

$$7) \sum \frac{4}{\sqrt{k} 2^k} (x-3)^k$$

$$\lim_{k \rightarrow \infty} \frac{4|x-3|^{k+1}}{\sqrt{k+1} 2^{k+1}} \cdot \frac{2^k \sqrt{k}}{4|x-3|^k} = \frac{|x-3|}{2} < 1 \Rightarrow 1 < x < 5 \quad R=2$$

$$\text{check } x=1 \quad \sum \frac{4(-1)^k}{\sqrt{k}} \text{ conv.}$$

$$x=5 \quad \sum \frac{4}{\sqrt{k}} \text{ div.}$$

$$8) f(x) = (1+x)^{\frac{1}{2}} \quad f(0) = 1$$

$$f'(x) = \frac{1}{2\sqrt{1+x}} \quad f'(0) = \frac{1}{2}$$

$$f''(x) = \frac{-1}{4(1+x)^{3/2}} \quad f''(0) = \frac{-1}{4}$$

$$1 + \frac{1}{2}x + \frac{-1/4}{2}x^2$$

$$9) u = \langle 0, 2, -3 \rangle$$

$$v = \langle 1, 0, -1 \rangle$$

$$a) \|u - 2v\| = \|\langle -2, 2, -1 \rangle\| = 3$$

$$b) u \times v = \begin{vmatrix} i & j & k \\ 0 & 2 & -3 \\ 1 & 0 & -1 \end{vmatrix} = \langle 2, -3, 2 \rangle$$

$$c) \text{No, } u \neq v \text{ b/c } u \cdot v = 3$$